PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			1
Title	1	Identify the report as a systematic review.	1
ABSTRACT			3
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	3
INTRODUCTION			4-5
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	4-5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	5
METHODS			5-8
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	5-6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	5
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	5
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g., for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	7
	10b	List and define all other variables for which data were sought (e.g., participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	7
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	7
Effect measures	12	Specify for each outcome the effect measure(s) (e.g., risk ratio, mean difference) used in the synthesis or presentation of results.	7
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g., tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	7
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	7
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	7
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	8
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g., subgroup analysis, meta-regression).	8
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	8
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	8
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	8

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Section and Topic	Item #	Checklist item	Location where item is reported
RESULTS			8-12
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	8
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	8
Study characteristics	17	Cite each included study and present its characteristics.	8-9
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	9
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimates and its precision (e.g., confidence/credible interval), ideally using structured tables or plots.	9
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	9-10
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g., confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	10
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	11
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	11
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	11
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	
DISCUSSION			12-14
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	12
	23b	Discuss any limitations of the evidence included in the review.	13
	23c	Discuss any limitations of the review processes used.	14
	23d	Discuss implications of the results for practice, policy, and future research.	14
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	5
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	15
Competing interests	26	Declare any competing interests of review authors.	15
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	15

Table S2 Search strategy used in current review. All searches were conducted on December 28, 2020.

The Cochrane Library	
#1	MeSH descriptor: [Azithromycin] explode all trees
#2	(azithromycin) OR (azithromicin) OR (azythromycin)
#3	#1 OR #2
#4	MeSH descriptor: [Bronchiolitis Obliterans] explode all trees
#5	("BO" OR "BOS" OR "ob") AND (bronchiolitis*)
#6	(bronchiolitis*) AND (constrictive or exudative or proliferative or obliterans)
#7	#4 OR #5 OR #6
#8	#3 AND #7
PubMed	
1	"Azithromycin"[Mesh]
2	(azithromycin) OR (azithromicin) OR (azythromycin)
3	("Azithromycin"[Mesh]) OR ((azithromycin) OR (azithromicin) OR (azythromycin))
4	Bronchiolitis Obliterans [MeSH Terms]
5	((bo) OR (bos) OR (ob)) AND (bronchiolitis*)
6	(bronchiolitis*) AND (constrictive OR exudative OR proliferative OR obliterans)
7	((Bronchiolitis Obliterans [MeSH Terms]) OR (((bo) OR (bos) OR (ob)) AND (bronchiolitis*))) OR ((bronchiolitis*) AND (constrictive OR exudative OR proliferative OR obliterans))
8	(("Azithromycin"[Mesh]) OR ((azithromycin) OR (azithromicin) OR (azythromycin))) AND (((Bronchiolitis Obliterans [MeSH Terms]) OR (((bo) OR (bos) OR (ob)) AND (bronchiolitis*))) OR ((bronchiolitis*) AND (constrictive OR exudative OR proliferative OR obliterans)))
Embase	
#1	'Azithromycin'/exp
#2	azithromycin OR azithromicin OR azythromycin
#3	#1 OR #2
#4	'Bronchiolitis Obliterans'/exp
#5	(bo OR bos OR ob) AND bronchiolitis*
#6	bronchiolitis* AND (constrictive or exudative or proliferative or obliterans)
#7	#4 OR #5 OR #6
#8	#3 AND #7
Web of science	
#1	TS=azithromycin OR TS=azithromicin OR TS=azythromycin
#2	TS=bronchiolitis*
#3	TS=bo OR TS=bos OR TS=ob
#4	TS=constrictive OR TS=exudative OR TS=proliferative OR TS=obliterans
#5	#3 AND #2
#6	#4 AND #2
#7	#6 OR #5
#8	#7 AND #1

Table S3 Assessment for study quality.

The Cochrane Risk of Bias Tool for Randomized Controlled Trials (RCT).						
Study	Sequence generation	Allocation concealment	Blinding participants and personnel	Incomplete outcome data	Selective reporting	Others（funding）
Lam, 2011	unclear	unclear	unclear	low	low	high

The Newcastle-Ottawa Scale (NOS) for Cohort studies.									
Study	Selection				Comparability	Outcome			total
	Representativeness of the exposed cohort	Selection of non-exposed cohort	Ascertainment of exposure	Demonstration that outcome of interest was not present at start of study	Comparability of cohorts on the basis of design or analysis	Assessment of outcome	Was follow-up long enough for outcomes to occur	Adequacy of follow up the cohorts	
Jain, 2010	1	1	1	1	2	1	1	1	9
Vos, 2010	1	1	1	1	1	1	1	1	8
Federica, 2011	1	1	1	1	1	1	1	1	8

Quality Appraisal Checklist for Case Series Studies											
study		Gerhardt, 2003	Verleden, 2004	Shitrit, 2005	Khalid, 2005	Yates, 2005	Verleden, 2006	Gottlieb, 2008	Porhownik, 2008	Norman, 2011	Williams, 2016
Study objective	Was the hypothesis/aim/objective of the study clearly stated?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Study design	Was the study conducted prospectively?	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	No	Yes
	Were the cases collected in more than one centre?	No	No	No	No	No	No	No	No	No	Yes
Study population	Were patients recruited consecutively?	Unclear	Unclear	Unclear	Unclear	Unclear	Unclear	Yes	Unclear	Unclear	Yes
	Were the characteristics of the patients included in the study described?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Were the eligibility criteria (i.e. inclusion and exclusion criteria) for entry into the study clearly stated	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Did patients enter the study at a similar point in the disease?	Yes	No	Yes	Yes	No	No	No	No	Yes	Yes
Intervention and co-intervention	Was the intervention of interest clearly described?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Were additional interventions (co-interventions) clearly described?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Outcome measure	Were relevant outcome measures established a priori?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Were outcome assessors blinded to the intervention that patients received?	Unclear	Unclear	Unclear	Unclear	Unclear	Unclear	Unclear	No	Unclear	Unclear
	Were the relevant outcomes measured using appropriate objective/subjective methods?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Were the relevant outcome measures made before and after the intervention?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Statistical analysis	Were the statistical tests used to assess the relevant outcomes appropriate?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Results and conclusions	Was follow-up long enough for important events and outcomes to occur?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Were losses to follow-up reported?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Did the study provided estimates of random variability in the data analysis of relevant outcomes?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Were the adverse events reported?	Yes	No	No	Yes	Yes	No	Yes	Yes	Yes	Yes
	Were the conclusions of the study supported by results?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Competing interests and sources of support	Were both competing interests and sources of support for the study reported?	No	No	No	No	Yes	Yes	No	No	Yes	Yes

Table S4 Sensitivity analyses.

(1). FEV1

Study omitted	Estimate	95% CI		P	Heterogeneity	
					I^2 (%)	*P*
Short term (≤12weeks)						
Verleden, 2004	0.13	0.07	0.20	0.00	0.00	0.75
Khalid, 2005	0.15	0.08	0.22	0.00	8.06	0.46
Yates, 2005	0.19	0.10	0.28	0.00	0.00	0.54
Verleden, 2006	0.15	0.08	0.21	0.00	2.90	0.4
Porhownik, 2008	0.14	0.07	0.21	0.00	2.29	0.4
Lam, 2011	0.15	0.09	0.22	0.00	0.00	0.43
Vos, 2010	0.16	0.09	0.24	0.00	8.03	0.36
Combined	0.15	0.09	0.21	0.00	0.00	0.46
Medium term (12-24weeks)						
Gerhardt, 2003	0.17	0.01	0.32	0.03	27.12	0.25
Verleden, 2004	0.18	-0.02	0.37	0.07	42.53	0.18
Porhownik, 2008	0.26	0.00	0.52	0.05	61.02	0.08
Vos, 2010	0.29	0.11	0.47	0.00	7.50	0.34
Combined	0.22	0.05	0.39	0.01	41.68	0.16
Long term (>24weeks)						
Verleden, 2004	-0.08	-0.55	0.38	0.72	87.21	0.01
Porhownik, 2008	0.10	-0.73	0.92	0.82	94.69	0.00
Vos, 2010	0.34	-0.01	0.70	0.06	60.74	0.11
Combined	0.11	-0.39	0.62	0.65	91.00	0.00

(2). FEV1(%) compared to baseline (at treatment initiation)

Study omitted	Estimate	95% CI		P	Heterogeneity	
					I^2 (%)	*P*
Short term (≤12weeks)						
Verleden, 2004	9.20	-0.54	18.94	0.06	86.80	0.00
Khalid, 2005	8.50	-0.55	17.54	0.07	84.25	0.00
Yates, 2005	9.89	-0.41	20.20	0.06	87.73	0.00
Verleden, 2006	9.90	-0.30	20.10	0.06	88.07	0.00
Gottlieb, 2008	13.50	5.58	21.43	0.00	71.57	0.01
Lam, 2011	12.99	2.23	23.75	0.02	88.34	0.00
Combined	10.65	1.63	19.67	0.02	86.40	0.00
Medium term (12-24weeks)						
Gerhardt, 2003	8.50	-16.65	33.66	0.51	93.37	0.00
Verleden, 2004	14.81	-28.82	58.45	0.51	81.26	0.02
Gottlieb, 2008	23.81	12.22	35.39	0.00	0.00	0.34
Combined	15.29	-8.61	39.19	0.21	89.86	0.00
Long term (>24weeks)						
Meloni, 2011	33.30	17.46	49.14	0.00	–	–
Verleden, 2004	-2.24	-7.76	3.28	0.00	–	–
Combined	14.72	-20.07	49.51	0.41	94.20	0.00

(3). Sensitivity analyses - FEV_1 % predicted

Study omitted	Estimate	95% CI		P	Heterogeneity	
					I^2 (%)	*P*
Short term (≤12weeks)						
Shitrit, 2005	3.70	1.62	5.77	0.00	0.00	0.99
Williams, 2016	2.08	-2.30	6.46	0.35	56.14	0.13
Vos, 2010	1.64	-2.91	6.19	0.48	37.26	0.21
Combined	2.99	0.70	5.27	0.01	14.77	0.31
Medium term (12-24weeks)						
Gerhardt, 2003	1.36	-0.73	3.44	0.20	0.00	0.89
Shitrit, 2005	5.35	-1.02	11.72	0.10	88.48	0.00
Norman, 2011	4.78	-2.32	11.87	0.19	89.13	0.00
Williams, 2016	4.81	-2.25	11.87	0.18	89.10	0.00
Vos, 2010	4.99	-2.59	12.57	0.20	88.08	0.00

Combined	4.19	-1.27	9.65	0.13	85.70	0.00
Long term (>24weeks)						
Shitrit, 2005	-7.90	-11.92	-3.88	0.00	–	–
Vos, 2010	-2.00	-7.64	3.64	0.00	–	–
Combined	-5.30	-11.04	0.45	0.07	64.17	0.09